

# SHARAD TEMBHURNE

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## WORK EXPERIENCE

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**Sr. Software Engineer** — *KPIT Technologies, Novi, Michigan*

**Jan 2024 - Present**

- Developed **Python** automation suite leveraging TestRail APIs, by utilizing AI-assisted development tools to update testcase data for **4,800+** test cases and reduced update time from **30 hours to under 10 minutes**.
- Automated a manual, repetitive workflow via **Python (PyWinAuto, PyAutoGUI)** to drive a legacy internal tool with no exposed APIs, cutting a routine 20–30 minute task to under 5 minutes (80-85% reduction)
- Designed and built an internal **AI agent** that analyzes test plans and recommends optimal test execution sequencing, reducing test cycle time by an average of 20% (ranging 12–33%) across varying plan sizes.
- Built a **Java** application to communicate with locally hosted non-Java services by developing **Python API wrappers** and using **socket programming** to enable reliable cross-language, on-system data exchange.
- Re-architected the core codebase and APIs of ECU Validator, an internal automotive test-execution tool, using **Java and Angular** to extend support from UDS to the J1939 protocol, enabling a second major automotive communication standard and securing over **\$480K in annual revenue**.
- Conducted in-vehicle testing to validate **Over-The-Air (OTA)** software update correctness and bug fixes across release cycles, **achieving a 100% on time delivery**.

**Software Engineer** — *KPIT Technologies, Pune, India*

**Nov 2019 - July 2022**

- Developed scripts in **TypeScript** to test software compliance with various communication protocols such as **UDS, J1939, KWP** standards across multiple automotive platforms.
- Automated TypeScript script generation using **Java**, reducing manual test preparation time by **95%** and accelerating test case coverage for automotive control units.
- Built an internal automotive test-execution tool (ECU Validator) using **Maven on Java** that ran diagnostic TypeScript sequences against vehicle modules.
- Upgraded the front-end features of ECU Validator in **Angular and React** improving the interface engineers use to run and review the diagnostic results.

## EDUCATION

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**New York University**

**Sept 2022 - Dec 2023**

Masters of Science in Computer Engineering - **GPA 3.92/4**

**Maharashtra Institute of Technology, Pune**

**Aug 2015 - May 2019**

Bachelors of Engineering in Electronics and Telecommunication - **CGPA 7.84/10**

## PROJECTS

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**Personal Portfolio Website**

**Oct 2023**

- Built a responsive single-page application using **Angular, React.js, CSS3, and Three.js**, incorporating 3D spline-based models for interactive graphics.

**Chess Visualization and Analytics**

**May 2023**

- Designed **RESTful APIs** for migrating chess game data from PGN format to **MongoDB**, optimizing database query efficiency by **40%** and built data analytics dashboards using Flask, JavaScript, and Jupyter Notebooks.

**Doraemon Gadget Web Catalog**

**Apr 2023**

- Implemented **infinite scrolling and quick search** functionalities in JavaScript and HTML5, with a mobile responsive interface that improved cross device navigation speed and user retention

## SKILLS

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**Programming Languages**

Python, Java, JavaScript/TypeScript, Linux, Unix

**Development**

React, HTML, CSS3, REST APIs, Flask, NodeJS, Angular, Three.JS

**Databases**

MongoDB, SQL, PostgreSQL, GraphQL, MySQL

**Tools**

Git, Docker, Jira, Confluence, AWS, Maven, Selenium, Postman, Jenkins

**AI**

GitHub Copilot, Cursor, Agentic AI